Tiffany Chu

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Education

University of British Columbia 2025 - Jul 2026
M.Sc. in Data Science
- Scholarship recipient, 94% average

University of British Columbia 2019 - Dec 2023
B.A. in Cognitive Systems, minor in Commerce
- Entrepreneurship@UBC accelerator venture program
- UBC Current Dragonboat Team - qualified for Nationals and Worlds Championships

Experience

Research Analyst Feb 2025 - Present
Lodestar Media Inc
- Identified reporting bottlenecks and automated recurring data collection and cleaning across multiple sources using Python, streamlining a previously manual weekly reporting process and improving data consistency for analysis
- Built a new PostgreSQL database in pgAdmin to merge and transform multiple raw tables, enabling analysis across 30,000+ company records, 20,000+ contacts, and historical executive and financial records for each company
- Communicated data insights to internal teams and served as a primary point of contact for client inquiries and support, triaged inquiries, troubleshot data issues, escalated, and documented fixes

Technical Research Assistant Sept 2024 - Jan 2025
Rural eMentoring BC
- Continued development of the NLP project, analyzed datasets of 10,000+ user entries, applying machine learning models to uncover behavioral insights and translate findings into reports and visualizations for non-technical stakeholders
- Enhanced an internal analysis tool by implementing 3 major new features to improve usability and performance

Data Science (NLP) Research Intern May - Aug 2024
UBC Data Science Institute
- Developed and deployed a multi-page Streamlit NLP application with modular pipelines for text cleaning, keyword filtering, sentiment/mood detection, summarization, topic modeling (LDA and BERTopic), and a recommender system
- Collaborated with cross-functional groups to define requirements and align product development with project objectives
- Conducted exploratory data analysis and applied time series analysis (using Prophet) to uncover and predict trends

Business Development Intern Sept - Dec 2022
Pawsome Concierge
- Conducted competitor and market analyses to identify partner opportunities and growth initiatives
- Designed a comprehensive client acquisition proposal outlining the pricing terms, value proposition, and timeline
- Enhanced marketing initiatives by translating market data, delivering insightful reports with statistics and visualizations

Byte Camp May - Aug 2020
Computer Programming Instructor

- Successfully taught computer applications such as Blender, OBS, Inkscape, and Godot (in Python) for weekly groups of 30 high school students alongside partner instructor

Volunteering and Mentoring Experience

Research Assistant 2023 - Dec 2024

Oculomotor Lab - Department of Visual Sciences, UBC

- Conducted experiments using the EyeLink eye tracker with participants, analyzed audio files in Python notebooks, created graphs and experiment design diagrams, and synthesized audio features into .csv spreadsheets
- Presented and conducted literature reviews to support research project ideation and methodology development

Patient Partner Jan 2023 - Present

UBC School of Nursing

- Supervisor: Dr. A. Fuchsia Howard
- Research Project - Analyzing transcripts from intensive care unit (ICU) survivors using computational methods to identify the challenges people experience after hospitalization that might contribute to an unplanned readmission
- Aided in the knowledge translation of a longitudinal Photovoice Study of ICU Survivors. Successful CIHR grant funding. Research paper and virtual exhibition in progress.

Founder and President 2020 - 2021

Unload Canada (UBC Chapter)

- Responsible for planning conferences with professors, and collaborated with other organizations (Kaleidoscope, Music for Mental Health, QPR training)

Volunteer Coding Instructor 2019 - 2020

The C.O.D.E Initiative

- Successfully trained a group of students in coding the Sphero SPRK+ robot by guiding students to think programmatically when solving errors in their code

Mentor 2020 - 2021

LEAP Canada

- A role model to incoming first-year students, collaborated with industry professionals and peers to deliver a presentation to an audience of over 200 students

Projects

Reproducible Framework for IBD Analysis with the BC Children's Hospital

- Engineered a reproducible R/Quarto pipeline and Shiny dashboard to analyze mycobiome, dietary, and clinical data for patients
- Applied statistical modeling (PERMANOVA, Spearman, PCA) to identify diet-microbiome associations, delivering results via an interactive dashboard and public analysis website

Latent Content Analysis of ICU Photovoice Transcripts with the UBC School of Nursing

- Created a pipeline using Python to analyze qualitative data, including CSV and Word-to-JSON converters, text preprocessing, embeddings/vectorization, and performance metrics (coherence, perplexity).
- Used topic modeling and other natural language processing techniques to identify themes in ICU patient transcripts and photos.

Anomaly Detection of Congressional Investments

- Identified congressional insider trading using data from the Senate Stock Watcher API.
- Implemented unsupervised ML ensemble methods (autoencoders and hierarchical clustering) to flag suspicious transactions and trading patterns.

Relevant Skills

Programming:	Python, VBA, Racket, R, HTML/CSS/Javascript, SQL(PostgreSQL, MongoDB, MySQL), Matlab, LaTeX
Technologies:	Docker, Git, AWS, Hadoop, Spark, Tableau, PowerBI, Streamlit, MS fabric, pgAdmin 4, DuckDB, Airflow, dbt, R Shiny, Quarto, Jupyter
Software:	Power Automate, Excel, MS Suite, G Suite, SPSS, Jamovi, NetLogo
Libraries:	pandas, NumPy, SciPy, scikit-learn, PyTorch, Hugging Face Transformers, NLTK, BERTopic, matplotlib, seaborn, Plotly, Altair, ggplot2, dplyr, yfinance
ML + statistical methods :	NLP, topic modeling (LDA, BERTopic), sentiment analysis, anomaly detection (autoencoders, hierarchical clustering), hypothesis testing (Kruskal-Wallis, Mann-Whitney, PERMANOVA), PCA, feature engineering, ETL/data pipelines, dashboarding
Certifications:	Red Cross Emergency First Aid AED+CPR certified